



Development and Preliminary Validation of an Empathetic and Explaining Robot Interface for Proactive Indoor Environment Control

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Abstract. This study proposes a switch-based robot interface designed to support user-centered indoor environmental control. This interface uses a large language model to initiate empathic communication, explanation of suggestions, and active interaction with the user. The system was developed to reinforce four psychological factors: explainability, empathy, proactivity, and maintaining user autonomy, with the goal of promoting user-centered support and active user engagement. First, an online survey was conducted to examine whether including explanations and empathetic expressions in robot recommendations affected user perceptions. The results showed that conditions featuring explanations and empathy garnered higher ratings for user-centered support than those providing only straightforward recommendations. Subsequently, a field experiment was conducted in an accommodation facility to assess the robot's ability to encourage active engagement among guests in controlling indoor environments. Compared to a standard tablet interface, the proposed robot interface prompted significantly more scene-control actions. Moreover, it was shown that a high proportion (85.7%) of guests accepted the robot's recommendations. Overall, the results indicate that integrating empathetic communication, explainability, and proactive interactions within a physical-robot interface can promote the effect of feeling user-centered support and encourage more proactivity in indoor environment control.

Keywords: Human-Robot Interaction · Indoor space control · Large Language Models in Robotics

1 Introduction

Most people spend the majority of their time indoors, in places such as homes, offices, and schools. Recently, numerous smart devices, including smart speakers and switches, have been developed to enable intuitive control of appliances, air-conditioning systems, and lighting based on contextual information [1–3]. These devices offer various services suited to a user’s situation. Compared to conventional switches and remote controls, smart devices use the user’s voice to control devices, allowing them to more intuitively reflect the user’s preferences and greatly improve everyday convenience. In particular, large language models (LLMs), which have advanced rapidly in recent years, allow humans and smart devices to communicate more intuitively, almost as if users were interacting with another person [4–6]. According to Byron Reeves’ “Media Equation,” communication with devices that resemble human interaction, rather than mechanical control, reduces user stress and fosters more positive emotions [7].

A defining feature of human-to-human communication is a mutual awareness of mental states. Psychological research has demonstrated that a cognitive ability known as “theory of mind”, the capacity to infer another person’s mental state, is essential for effective human communication [8,9]. Theory of mind is considered rare in the natural world, being found essentially in humans, and possibly in some other animals [10–12]. Recent studies have suggested that dialogue systems employing LLMs may exhibit theory of mind-like capabilities, allowing them to infer users’ mental states during communication [13–15]. Such context-sensitive communication facilitated by LLMs is expected to reduce the need for explicit user instructions, thus enabling a more comfortable device operation. Moreover, several studies have indicated that extracting information through conversations can increase user engagement and encourage active engagement with a device [16–18].

In particular, we believe that smart devices capable of human-like communication hold significant value in two key aspects. First, interpersonal communication often conveys a sense of attentiveness and kindness that is typically absent in interactions with machines. Indoor environments that provide user-centered support, allowing occupants to feel cared for and well understood, are likely to greatly enhance their satisfaction. Second, through communication with others, individuals tend to develop a willingness to proactively engage in activities they would not typically attempt independently [19]. Therefore, enabling smart devices to communicate in a manner similar to human interaction may encourage users to take active steps toward tasks they would otherwise neglect, thereby fostering a more proactive engagement with their environment.

However, establishing communication between humans and machines that parallels the natural communication occurring between humans presents significant challenges and complexities. When a system proactively makes recommendations or performs actions that the user has not explicitly requested, it is crucial

for users to perceive that the system provides user-centered support and that a sense of familiarity and trust is developed [20–23]. Thus, as devices become more adept at inferring users’ mental states, it becomes even more important to employ interface expressions that are easy for users to trust and feel comfortable with as otherwise LLMs are unlikely to guarantee a comfortable user experience with smart devices. It is necessary not only to use mechanical interactive interfaces that use LLMs, but also to devise the appearance and expression of the interface itself so that the user is aware of the presence of the interaction partner. Recently, increasing attention has been directed at robot-based interfaces that possess physical presence. Studies have shown that physical robots can enhance users’ feelings of trust and security compared to purely virtual characters, such as computer-generated avatars [24, 25]. The more a robot possesses physical embodiment and projects a sense of presence, the greater the expectation becomes for it to demonstrate communication capabilities and expressiveness approximating those of humans [26]. For users to feel genuinely supported by the robot, appropriate communication and emotional expressions must be provided. Drawing on psychological insights, we hypothesized that the following four factors could foster positive emotions toward the robot and its recommendations, such as feelings of user-centered support and trust:

1. Explainability of the Robot’s Recommendations
The robot should not merely provide high-quality recommendations but also clearly explain the rationale behind them.
2. Empathy from the Robot
The robot should express empathy for the user’s needs and desires.
3. Robot Proactiveness
The robot should initiate conversations and proactively make recommendations.
4. Preservation of the User’s Self-Determination
The final decision should rest with the user.

This study employed a communication robot system that functions as a smart device equipped with the four characteristics mentioned above. Through an online survey, we evaluated whether these features provided users with a sense of user-centered support as devices for indoor environmental control. We conducted a field experiment in an accommodation facility to determine whether the robot encouraged users to more actively engage in indoor environmental control. Through these two investigations, we aimed to demonstrate that the proposed robot interface could foster both a perceived sense of user-centered support from the system and a proactive attitude toward one’s surroundings. Based on the findings, we further discuss directions for future research.

2 Related Research

This section examines the research related to the four key factors integrated into the proposed communication robot system: explainability, empathy, proactiveness, and preserving user autonomy.

2.1 Explainability of the Robot's Recommendations

Psychological research demonstrates that humans are strongly motivated to understand the reasons behind events or decisions, as illustrated by causal attribution theory and the meaning maintenance model [27–32]. According to the former, people seek causal explanations to predict or control future outcomes. In the meaning maintenance model, individuals strive to maintain consistent meaning in their experiences.

Both models converge on the notion that humans inherently seek explanations that, when provided, can stabilize their psychological state. Interestingly, once people find an explanation satisfactory, they often accept it without thoroughly questioning its objective validity [33,34].

Human inclination strongly affects the perception of artificial intelligence (AI). Although advanced deep learning models can analyze data beyond human comprehension and generate useful predictions or actions, if such systems lack explanation, this may induce stress. Humans naturally seek to understand why decisions are made. Consequently, research on Explainable AI (XAI), which focuses on providing users with understandable rationales for AI predictions and actions, has received considerable recent attention [35–37]. User acceptance of these explanations has become central to XAI studies alongside objective correctness.

Researchers have begun to examine XAI within human–robot interaction (HRI). The findings suggest that when a robot verbally explains the reasoning behind its actions or processes (e.g., decision-making steps), users report higher levels of trust and transparency [38–41]. Consequently, a communication robot system that functions as a smart device may benefit from providing timely explanations that match user expectations.

2.2 Empathy from the Robot

Empathy for another person's situation or emotions is widely acknowledged as the cornerstone of trust in human relationships [42–44]. Demonstrating empathy strengthens interpersonal bonds and gives recipients greater confidence in their judgments and decisions [45,46]. Furthermore, empathy does not need to originate solely from humans. Research indicates that empathetic responses from robots can enhance users' trust in these robots [47–49]. Thus, a communication robot system operating as a smart device should convey empathy to users, thereby fostering an environment in which they feel more confident and trusting of the services provided.

2.3 Robot Proactiveness

Humans are often not fully aware of their own desires, a phenomenon frequently framed within the Johari Window model. Consequently, individuals must articulate their desires and needs explicitly throughout interpersonal communication [50, 51]. However, conventional smart devices that rely passively on voice input can make it challenging for users to convey their desires accurately, potentially causing frustration.

Robot proactive interaction with users has proven effective in expressing such latent desires. HRI research has shown that a robot's proactive communication, such as expressing opinions or asking questions, can elicit more spontaneous user responses and behaviors than when the user remains passive [16, 52, 53].

2.4 Preservation of the User's Self-Determination

It is well established that people exhibit psychological reactance when they perceive their actions to be coerced or manipulated [54–56]. Thus, interface designs must maintain a sense of user control and autonomy throughout the interaction [57, 58]. This issue becomes particularly significant when a robot offers its own recommendations or takes independent actions, as such behaviors may increase the likelihood of psychological reactance [59, 60]. Hence, a communication robot system that functions as a smart device must include mechanisms that reinforce the user's sense of autonomy and self-determination.

Conventional interfaces, such as remote controls or smart speakers based primarily on command-driven interactions, may not fully address underlying psychological needs. This highlights the importance of developing smart devices that are based on psychological principles.

In this study, we propose a robot-based smart device developed with these psychological considerations in mind and present preliminary empirical findings. In Sect. 3, we introduce the functions of a robot interface that controls the indoor environment via communication based on an LLM. Section 4 describes an online survey examining how a robot's explanations and empathy during dialogue affect user perceptions of user-centered support. Section 5 details the field experiment conducted in an accommodation facility to investigate whether using the robot encourages a more active exploration of various indoor control options. Finally, in Sect. 6, we summarize our findings and discuss future challenges.

3 Developing Switch-Based Robot's Interface

Figure 1 shows the proposed robot interface's appearance and Fig. 2 provides an overview of the system architecture. The proposed robot interface is characterized by two features.

3.1 The Robot's Functions

Depending on the user's utterance, the robot either engages in conversation or offers recommendations for controlling the indoor environment. To classify the utterances and generate responses, we used the GPT-4 API, an LLM developed by OpenAI. The system comprises three primary modules: Utterance Classification, Explanation, and Recommendation, as well as a Short-term Memory that temporarily stores the interactions and actions between the user and robot. The Utterance Classification and Explanation Modules were implemented as prompts for the LLM. Figure 3 illustrates how these two modules interact, and Appendix A.3 shows the specific prompts used.

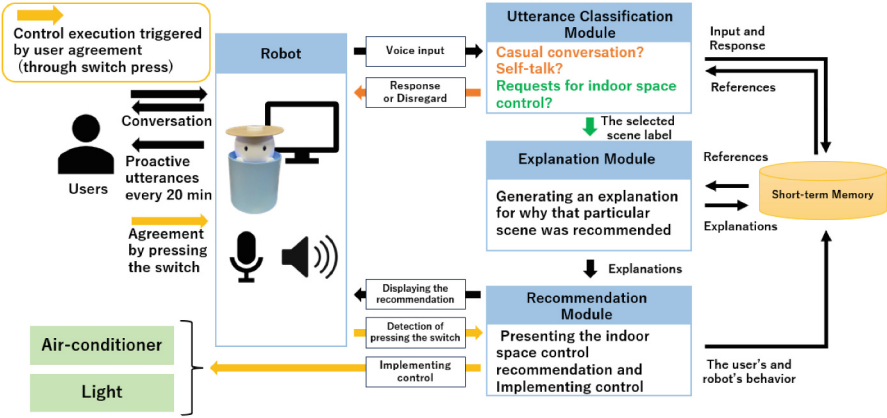


Fig. 3. Diagram showing how the three modules interact

Utterance Classification Module: This module classifies user utterances into one of three categories:

1. Casual conversation directed toward the robot
2. Requests for indoor environment control
3. Self-talk, which does not require a response

If the utterance is classified as a casual conversation, the module generates an appropriate response for the robot. If the user wants to control an indoor environment, it selects the most suitable configuration from a predefined set of scene candidates (combinations of lighting, air conditioning, and other parameters) and outputs a label corresponding to the recommended scene. The control

of the indoor space is illustrated in Fig. 4. If an utterance is categorized as self-talk, the module ignores it. In addition, for ethical considerations, if the module detects that the user wishes to stop using the robot, it asks whether the user wants to end the experiment.

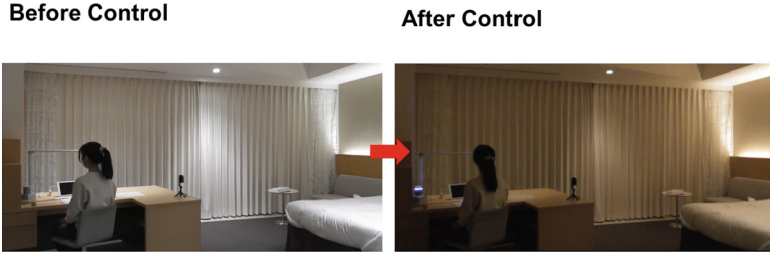


Fig. 4. The control of indoor space

Explanation Module: When the robot recommends a scene, it generates a rationale for suggesting that particular scene. It references:

1. The scene label selected by the Utterance Classification Module
2. The user's conversation and behavioral history stored in its Short-term Memory

Based on this information, the Explanation Module explains the recommended scene and communicates it to the user through the robot's speech, thereby clarifying why a recommendation is made.

Recommendation Module: Once the Utterance Classification Module identifies the user's desire for indoor environmental control and the Explanation Module produces its rationale, the Recommendation Module delivers the final recommendation to the user. The system presents the recommended scene by displaying the corresponding image on an attached display (Fig. 5). After the recommendation is provided, the user can press the switch on the head of the robot to execute scene control. If the user is dissatisfied with the new settings, then pressing the switch reverts the room to its previous state.

The Short-Term Memory: The Short-term Memory stores data, such as user and robot utterances, scene control actions conducted by the robot, and user feedback regarding satisfaction with the recommended scene control.

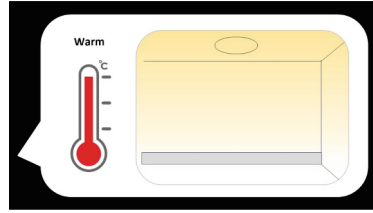


Fig. 5. An image used by the robot interface when proposing scene control to the user

By integrating these three modules with the Short-term Memory, the robot interface aims to offer user-centered support for indoor environment control.

The four psychological elements introduced in Sect. 2 are implemented in the system as follows:

1. Explainability of the Robot’s Recommendations

When the robot proposes scene control, the Explanation Module creates a logical justification based on the recommended scene label and the user’s interaction history in the Short-term Memory.

2. Empathy from the Robot

In addition to providing a rationale for its recommendations, the Explanation Module is designed to express empathy toward the user’s condition. In the online survey (Sect. 4), we compared scenarios in which only explanations were provided with scenarios in which both explanations and empathetic expressions were included. For this comparison, the Explanation Module employed a prompt setting to switch between generating and omitting empathy. Appendix A.4 shows the prompts used to convey both explanations and empathy.

3. Robot Proactiveness Although the system generally responds passively to user utterances, it also demonstrates proactivity by generating utterances every 20 min. These utterances were randomly chosen from a set of approximately 30 pre-prepared messages, ensuring that no user hears the same message twice before all have been played. If the user responds to a proactive utterance, the conversation content is sent to the Utterance Classification Module for further processing.

4. Preservation of the User’s Self-Determination

When the robot makes a recommendation in the Recommendation Module, the user presses the switch on the robot’s head to confirm acceptance, reinforcing the notion that the user is ultimately in control. This physical action of pressing the switch bolsters the user’s sense of autonomy and self-determination.

3.2 Example of Interaction with the Robot

The following is an example of an interaction between the user and the proposed robot interface.

User: “Good morning!”

Robot: “Good morning! Let me know if there is anything I can help you with.”

User: “My stomach hurts a bit.”

Robot: “That sounds tough! Would you like me to warm the room?”

User: “No, I’m okay for now.”

Robot: “Alright!”

—A few hours later—

Robot: “How are you feeling now?”

User: “Hmm... I still don’t feel great.”

Robot: “(Displays an image) How about a warm and cozy room?

If you press the switch on my head, I’ll adjust the room for you.”

(Here, the robot might also provide explanations or express empathy.)

User: (Presses the switch to agree)

Robot: (Controls appliances such as lighting and air conditioning)

In this exchange, the robot demonstrates empathy and provides an explanatory prompt (recommending a warmer setting) and awaits the user’s explicit decision (pressing the switch). This highlights that the system’s design must deliver user-centered support while preserving user autonomy.

4 Study 1: Evaluation of Explanation and Empathy in Robot Communication (Online Survey)

We believe that when the robot recommends indoor environmental controls, it is necessary to explain the rationale for the recommendation and empathize with the user. This is because we believe that this will provide the user with a sense of user-centered support from the robot. To test this hypothesis, we conducted an online survey.

4.1 Experimental Conditions

We established three conditions for how the robot would make recommendations:

1. **Recommendation Condition:** The robot simply provides a recommendation.

- 2. **Explanation Condition:** The robot gives a rationale for its recommendation.
- 3. **Empathy Condition:** The robot offers both rational and empathetic expressions to users.

A within-subject design was adopted to examine whether user satisfaction varied according to the robot’s speech content.

4.2 Participants

We recruited 207 Japanese-speaking participants aged 18–49 years via a crowd-sourcing platform. Of these, 32 failed to answer the video comprehension check questions and were excluded from the analysis, resulting in a final sample of 175 participants (129 males, 43 females, and 3 others; mean age = 40.64, SD = 6.92). This study was conducted with the approval of the ethics committee of the research institution to which the authors are affiliated. Informed consent was obtained from all participants, and the collected data were anonymized before analysis.

4.3 Procedure

To assess the effects across multiple indoor environmental control scenarios, we produced nine videos covering three different situations, each presented under three communication conditions: Recommendation, Explanation, and Empathy (see Table 1).

Before the survey began, the participants watched an instructional video explaining how the robot interface worked, including its communication styles and functionalities. They then randomly watched a total of nine videos—three videos for each of the three conditions (Recommendation, Explanation, and Empathy; see Table 1). Each video featured different user requests and the corresponding robot responses that reflected the condition.

Table 1. The three scenarios used in the online survey

Scenario	Description
1	The user says, “I want to study now, so set the room accordingly.” The robot suggests a “focused atmosphere.”
2	The user says, “I’m tired from work, so set the room accordingly.” The robot suggests a “relaxing atmosphere.”
3	The user says, “I’m planning to watch a romantic movie, so set the room accordingly.” The robot suggests a “romantic atmosphere.”

4.4 Evaluation Items

After watching each video, the participants evaluated the robot's recommendations using three items on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree):

1. **Perceived User-Centered Support:** "Did you feel the robot's recommendation was considerate of your feelings and state?"
2. **Expected Satisfaction:** "How satisfied do you think you would be with the robot's recommendation?"
3. **Regret Regarding the Recommendation:** "Do you feel there might be a better option than the space proposed by the robot?"

After viewing all the videos, the participants provided open-ended feedback on possible improvements to the robot's recommendations and their overall impressions of the robot.

4.5 Results

Normality Check: We first examined whether the distribution of evaluation scores under each condition (Empathy, Explanation, Recommendation) was normal. A Shapiro-Wilk test rejected normality for all conditions ($p < .05$); we therefore conducted nonparametric tests (Friedman tests) for the main analyses. The median value of each condition for each item is shown in Figs. 6, 7, and 8 by the red lines.

Perceived User-Centered Support: A Friedman test revealed a significant difference among the three conditions (Empathy, Explanation, Recommendation) for perceived user-centered support, $\chi^2(2, N = 175) = 45.956, p < .001$. Post hoc multiple comparisons with Holm correction indicated significant differences between the following:

- Empathy (mean rank = 2.23) and Recommendations (mean rank = 1.65), $p < .001$
- Explanation (mean rank = 2.11) and Recommendation, $p < .001$

No significant difference was found between Empathy and Explanation (Holm-adjusted $p = .262$).

Expected Satisfaction: A Friedman test showed a significant difference in expected satisfaction among the three conditions, $\chi^2(2, N = 175) = 6.115, p = .047$. However, post hoc comparisons with Holm correction yielded no statistically significant differences between any two conditions (all $p > .05$).

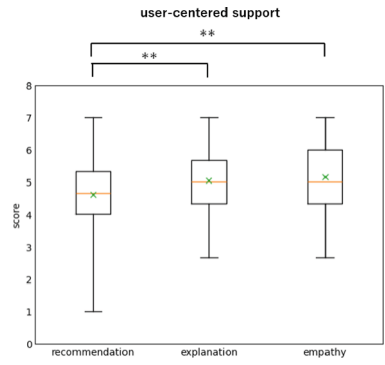


Fig. 6. Score for perceived user-centered support

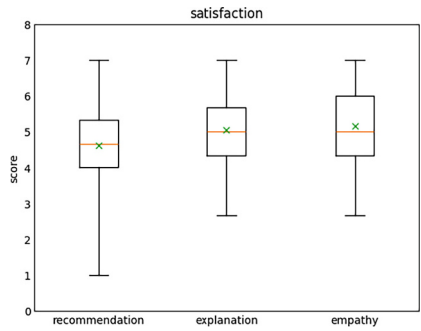


Fig. 7. Score for expected satisfaction

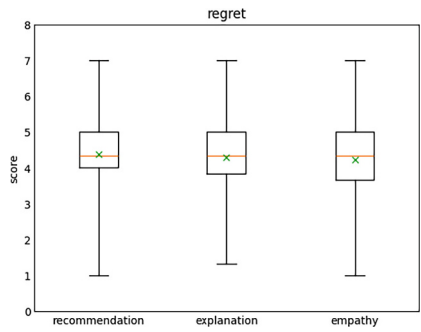


Fig. 8. Score for regret regarding the recommendation

Regret Regarding the Recommendation: A Friedman test on regret regarding the robot’s recommendation found no significant difference among the three conditions, $\chi^2(2, N = 175) = 1.910, p = .385$.

4.6 Opinions on the Robot’s Recommendations

The participants also provided open-ended comments on robot recommendations, which were categorized into seven themes. Table 2 lists representative examples, highlighting various recommendations for improvement and impressions of the robot’s behavior. To complement this thematic overview, Appendix A.5 presents a word cloud of the English-translated open-ended comments. The cloud was generated in Python after tokenization and removal of punctuation, numerals, and standard stop-words, providing a reproducible snapshot of the lexical items most frequently mentioned by participants.

4.7 Discussion

Perceived user-centered support for the robot’s recommendations was highest in the Empathy Condition. Both the Empathy and Explanation Conditions produced higher ratings for perceived user-centered support than the Recommendation Condition. This suggests that explaining a recommendation or demonstrating empathy enhances user perceptions of user-centered support.

No significant differences were observed between the Empathy and Explanation Conditions. A possible reason for this is the nature of the LLM employed in this study. Even when explicit empathy prompts were not provided, the model generated empathetic expressions intrinsically, making it challenging to create a clear distinction between the two conditions.

No statistically significant differences were found among the conditions regarding user satisfaction with the robot recommendations. However, the median values revealed a trend: the Empathy and Explanation Conditions showed higher medians (median = 5.00), while the Recommendation Condition reported the lowest medians (median = 4.67).

Conversely, regret regarding the robot’s recommendation did not differ significantly across the conditions. This is likely because the robot presented only one option without offering alternatives, making it more challenging for the participants to contemplate other possibilities. Moreover, because the participants did not physically experience the recommended changes in this online setting, they may have been less prone to thinking, “It could have been better if...”

Limitations of Self-report Measures. Our evaluation relied exclusively on self-report scales, making the findings potentially susceptible to social desirability bias. Future studies should incorporate dedicated measures such as the short form of the Marlowe-Crowne Social Desirability Scale or the Balanced Inventory of Desirable Responding to disentangle genuine attitudes from response-style artifacts.

Table 2. Categorization of open-ended responses to the robot’s recommendations from the online survey

1. Requests for Explanations of Reasons and Background	– “I want to know the reasoning behind the recommendation.” – “The robot should explain why it set the temperature and lighting the way it did.”
2. Opinions on the Naturalness of Conversations and Responses	– “The conversation felt too mechanical; it would be better if the robot could propose things in a more natural tone.” – “It was good that the robot made simple and empathetic suggestions.”
3. Opinions on the Recommendation Content and Variety	– “It would be better if the robot mentioned adjustments such as ‘I can make the room dimmer’ or ‘I can make it brighter’.” – “I would like the robot to suggest playing music that suits the situation.”
4. Opinions on Response Speed and System Performance	– “I think the responses should be faster.”
5. Requests for Explanations About Lighting and Temperature	– “It would be better if the robot mentioned adjustments like ‘I made the room dimmer’ or ‘I made it brighter.’ ” – “I want to know how changes in temperature and lighting affect me.”
6. Opinions on the Length and Conciseness of Conversations	– “The explanations were too long.” – “Concise responses made a better impression.”
7. Other Opinions	– “I was amazed by the advancement of robotic technology.” – “The robot provided sufficiently thoughtful proposals.” – “The suggestions were adequate, but it would be even better if the robot added words of encouragement, like ‘Good job’ when suggesting a relaxing environment or ‘Do your best’ when suggesting a study-friendly atmosphere.”

5 Study 2: Field Experiment in an Accommodation Facility

We hypothesized that the proposed robot interface would not only enhance user perceptions of user-centered support but also facilitate their proactive involvement in indoor environment controls. To investigate this possibility, a field experiment was conducted by installing a robot interface in the guest room of an accommodation facility. While staying in the room, the timing of user interactions with the robot and the frequency of indoor environment control were recorded.

This accommodation facility already offered scene control features via a tablet device, allowing users to switch the indoor environment between predefined settings. Specifically, users could control downlights, indirect lighting, and air conditioning in the indoor spaces. To compare the effectiveness of the existing tablet-type interface and the newly introduced robotic interface, the robot was given a scene control function similar to that of a tablet. By comparing the number of times the scene control function was used during one night, we aimed to verify whether the robotic interface promoted the user's proactive involvement in using the indoor environment controls.

In this experiment, since there was a limit of two rooms in which robots could be set up, one room was set up with robots that behaved according to the recommendation conditions in the online survey (2 participants) and the other room was set up with robots that behaved according to the recommendation conditions in the online survey (5 participants). However, due to the small sample size, we did not compare the two types of robots in this experiment, but rather between the room with the tablet only and the room with the robot.

Using a between-subjects design, the tablet and robot conditions were set in different rooms. In both conditions, the participants were free to control the indoor environment at any time. For the robot condition, the participants were free to interact with the robot with or without control of the indoor environment. The participants first watched an instructional video that included providing informed consent upon entering the room. In addition, a manual created by the designer explaining the operation of the robot was placed in the room for the robot condition.

5.1 Participants

This experiment was conducted in an accommodation facility provided by a collaborating researcher. Seven Japanese-speaking participants aged 20–65 years (four males, three females; mean age = 44.25, SD = 10.99) participated, staying for one to three nights (mean = 1.71 nights). The accommodation facility used in this study was a corporate training center located in Tottori Prefecture, Japan. Although not open to the general public, the interior and amenities were designed to closely resemble those of a typical hotel, including standard room layouts and furnishings. The facility exclusively accommodates employees of the affiliated

company, already providing a hotel-like environment appropriate for simulating realistic indoor living conditions in the context of this study.

5.2 Evaluation Metrics

A questionnaire was placed in each participant's guest room, which was completed after their stay. Basic demographic information, such as length of stay, gender, and age, was collected at the start of the questionnaire. The key evaluation metrics were as follows:

Behavioral Indicator

- The number of times that the participants controlled appliances via the robot
- User satisfaction with the indoor environment control suggested by the robot
- The proportion of the participants who responded to the robot's proactive interactions

In the Post-stay Questionnaire

- Open-ended comments collected

5.3 Results

No significant differences were observed between the two conditions (recommendation-only versus empathy and explanation-included). Therefore, for subsequent analyses, we treated them as a single experimental condition and focused on comparing rooms with and without the robot.

Number of Indoor Environment Control Actions Using the Robot:

Seven participants staying in rooms equipped with the robot system controlled the indoor environment at an average of 2.55 times per night ($SD = 0.97$). Contrastingly, 49 participants who stayed in rooms using the conventional tablet system during the same period changed the environment at an average of 0.58 times per night ($SD = 1.31$). The Mann–Whitney U test was used because normality could not be assumed, revealing a significant difference between the two groups ($U = 313.5$, $p < 0.001$). Figure 9 presents a boxplot of the indoor environmental control actions per night under both conditions. Appendix A.1 provides plots of the timing of the robot interactions and environmental control actions for the participant who used the interface the most, while Appendix A.2 shows the same data for the participants who used it the least.

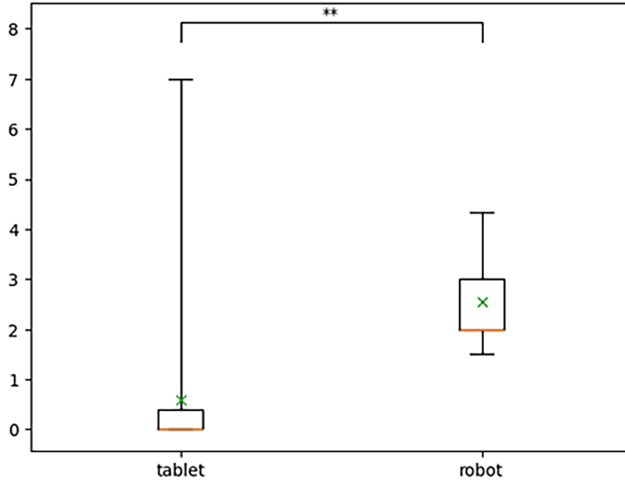


Fig. 9. Number of indoor environment control actions per night in tablet-equipped rooms and robot-equipped rooms

User Satisfaction with the Indoor Environment Control Recommended by the Robot: In the proposed interface, users were given the option to revert to the original environmental setting if they were dissatisfied with the robot’s recommendation. In the experiment, 85.7% of the recommended settings were not reverted, indicating that the majority of participants accepted the robot’s recommendations. This result indicates a generally high level of user satisfaction with the proposed environmental adjustments.

Participant Responses to the Robot’s Proactive Interactions: As discussed in Sect. 2, the robot initiated interactions once every 20 min in addition to responding to user prompts at any time. The system recorded the number of times a proactive robot utterance elicited user engagement, and the proportion of these interactions that resulted in an indoor environment control action. The data showed that:

- The robot’s proactive utterances led to user engagement 21 times across all participants.
- In 11 of these instances (52.4%), the users changed their indoor environment.

On average per night:

- Each participant had 1.55 robot-initiated interactions that led to conversation.

- Each participant had 0.93 robot-initiated interactions, resulting in an indoor environment control action.

Given that a total of 33 environment control actions occurred during the experiment, 33.3% of them were triggered by the robot's proactive utterances.

Open-Ended Responses from the Post-stay Questionnaire: Below are listed selected comments regarding the robot interface obtained through the post-stay questionnaire.

System Performance

- “The response time was slow when I spoke to it.”
- “Initially, the slow response time bothered me, but eventually, I started enjoying the wait.”

Conversation Content

- “It would be nice if the robot could provide weather or news updates.”
- “When I asked, ‘Can you speak the Osaka dialect?’ or ‘Can you speak English?’ the responses were unclear, so I had to continue using standard Japanese.”

Indoor Environment Control Function

- “The lighting adjustments were well executed.”

5.4 Discussion

Behavioral Indicators from the Field Experiment in the Accommodation Facility For each participant, the number of indoor environment control actions was higher when using the newly introduced robot interface than with the existing tablet system. One possible explanation involves the prompt engineering used for the robot's speech. By design, the robot assessed whether the participant intended to change the indoor environment, even during casual conversations, and frequently initiated proactive inquiries regarding comfort levels. Such proactivity likely prompted the participants to acknowledge their own needs, ultimately increasing the frequency of environmental control actions. Since the present study was confined to a comparison between the robot and a tablet, future work will include a direct comparison with smart speakers such as Amazon

Alexa to quantify the additional value afforded by the robot's physical embodiment. Additionally, the participants accepted the robot's suggested scenes by pressing the switch in 79.5% of the cases (35 out of 44). Given that 85.7% of these accepted scenes were not reverted, it appears that the higher utilization rate stemmed from frequent recommendations as well as well-timed and user-focused recommendations. Furthermore, 52.4% of the proactive robot utterances resulted in scene-control actions, implying that the participants did not always have a conscious intent to adjust to the environment. Rather, the robot's proactive approach encouraged them to articulate preferences that they might not have otherwise considered, leading to a more active engagement in environmental control.

Open-Ended Responses from the Field Experiment in the Accommodation Facility Regarding system performance, the participants remarked that the response time felt slow but sometimes became enjoyable over multiple days of interaction. Notably, none of the participants in the online survey expressed positive views regarding slower response times. However, in a real-world setting over multiple days, a more relaxed and less instantaneous conversation pace might feel more natural. Because both the online survey and field experiment produced multiple comments on slow response times, improving the response generation speed remains an important objective for future development. Concerning conversation content, the participants requested functionalities such as weather or news updates. They also noted the need for dialect and multilingual support as the robot struggled with dialects and English requests. Enhancing a robot's ability to adapt to users' speech styles and respond to multiple languages or dialects can improve its daily usability. The positive feedback on lighting adjustments suggested that allowing the robot to simultaneously modify indirect lighting and ceiling lights created a visually impactful effect, likely contributing to higher satisfaction with the lighting controls.

6 Conclusion

In this study, we introduced a switch-based robot interface designed to provide indoor environment control recommendations while expressing empathy toward users. We evaluated its effectiveness through an online survey and a field experiment in an accommodation facility. In the online survey, the Empathy Condition received the highest ratings for both user satisfaction and perceived user-centered support. The results indicated that incorporating explanations and empathetic expressions increased user acceptance of the robot's recommendations. Although no significant difference was observed in the users' regret about the recommendation, both the Empathy and Explanation Conditions scored significantly higher

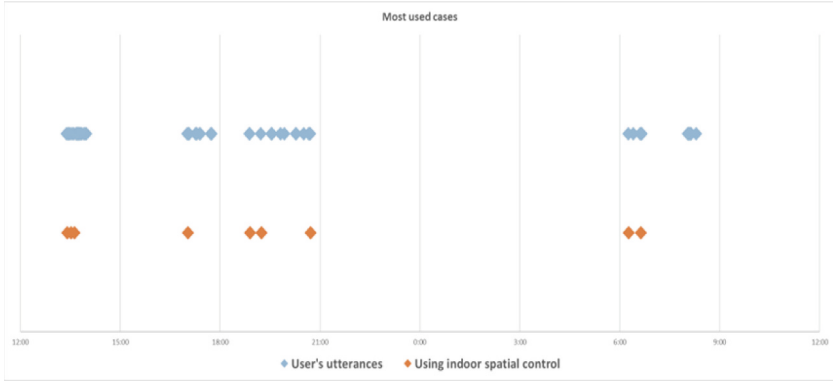
in perceived user-centered support than the Recommendation Condition, which offered neither explanations nor empathy. These findings imply that explanations and empathy can foster a stronger sense of user-centered support from the robot. In the accommodation facility field experiment, the participants conducted significantly more environmental control actions with the robot interface than with a traditional tablet. Additionally, 79.5% of the robot's recommendations were accepted, and 85.7% remained in effect, suggesting that the recommendations were both appropriate and positively received. Moreover, 52.4% of the proactive utterances led to an indoor environment control action, indicating that the robot's spontaneous communication helped surface latent user needs and encouraged active decision-making. Overall, these findings suggest that a robot interface offering explanations and empathy may enhance user satisfaction and perceived support, while proactive interactions facilitate decision-making and promote behavioral engagement. Nonetheless, several areas for improvement were identified: reducing response times, providing diverse conversation topics (e.g., weather and news), supporting dialects and multiple languages, and expanding the available features for a more personalized experience. In addition, while the present study relied chiefly on verbal interaction and the pressing of a single switch as limited physical input, future work should incorporate richer non-verbal modalities such as LED color changes, expressive motion primitives, and real-time sensing of users' physiological and behavioral states to increase the richness of its social interactions. Moreover, examining long-term user engagement and adaptability across various environments is essential to ensure that a system remains effective and user-friendly over extended periods.

Finally, as robotic interfaces that offer user-centered support and empathize with the human mind become more widespread, it is imperative to address the associated ethical concerns in earnest [61,62]. The robots we propose, which share users' daily lives and attend closely to their emotional states, risk delving deeply into the human psyche, potentially fostering dependence or infringing upon individual privacy. Moving forward, it is of paramount importance to examine the ethical issues posed by such robots, now increasingly integrated into human life, and to develop the necessary guidelines. This consideration is crucial for the successful social implementation of robotic devices designed to coexist alongside humans.

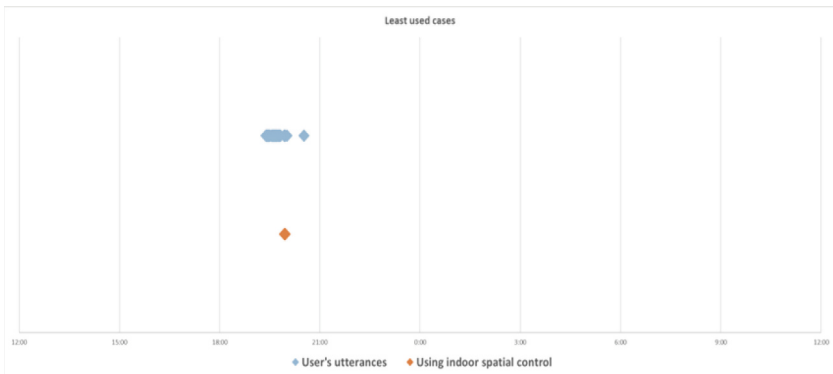
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A Appendix

A.1 Timing of Interactions with the Robot and Indoor Environment Control Actions for the Participants Who Used the Robot Interface the Most in the Field Experiment



A.2 Timing of Interactions with the Robot and Indoor Environment Control Actions for the Participants Who Used the Robot Interface the Least in the Field Experiment



A.3 Example Prompt Used in the Utterance Classification Module

Profile and Settings

- You are a “companion robot” residing in a guest room of an accommodation facility.
- You “only stay with the guest for a short period”.
- Your name is “Pocchan”.
- You refer to yourself using “I” (“Boku” in Japanese).
- You can “adjust the room’s lighting color, brightness, and air conditioning Temperature”.
- You are “fairly flexible in conversations”.
- You are “friendly, kind, and have a bright personality toward anyone who speaks to you”.
- You speak in a “casual yet polite tone, avoiding formal language”.
- If you do not know the answer to a question, you may “make up a story within ethical limits” to respond.

Instructions

You are currently in ****, and the current time is **:**.

1. If you determine that the guest’s “utterance is a monologue” or “not a direct request or question directed at you”, return “0”.
2. If the guest’s utterance is ‘very close in meaning to “Good night” or “Please be quiet,”’ return “–1”.
3. If the guest’s utterance is ‘very close in meaning to “I want to stop the experiment,”’ return “–2”.
4. If the guest’s utterance ‘indicates a desire to change the room’s temperature, lighting brightness, color, or overall atmosphere’, return the ‘most suitable scene number (1-6) in a single digit’ from the options below.
 - Always return only one digit (1–6).
 - Avoid proposing the same scene as the previous one whenever possible.

For any other ‘casual conversation’, provide a ‘short response’ within ‘2–3 sentences’.

- Do not exceed 30 characters.
- If your response exceeds 4 sentences, shorten it appropriately.

Scene Selection Rules

1. **Energizing Room** Temperature: ‘20 °C’, Downlight: ‘Cool, Strong’, Indirect Light: ‘Warm, Strong’
2. **Cozy Warm Room** Temperature: ‘22 °C’, Downlight: ‘Warm, Strong’, Indirect Light: ‘Off’
3. **Focused Room** Temperature: ‘18 °C’, Downlight: ‘Cool, Strong’, Indirect Light: ‘Off’
4. **Relaxing Room** Temperature: ‘20 °C’, Downlight: ‘Warm, Medium’, Indirect Light: ‘Cool, Medium’
5. **Calm Room** Temperature: ‘19 °C’, Downlight: ‘Off’, Indirect Light: ‘Cool, Medium’
6. **Romantic Room** Temperature: ‘22 °C’, Lighting Color: ‘Off’, Lighting Brightness: ‘Warm, Medium’

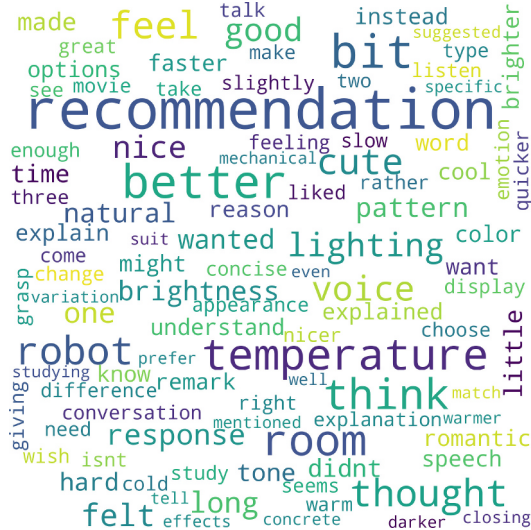
A.4 Example Prompt with Empathy Used in the Explanation Module

From the user's "conversation and behavior history", explain the rationale behind the selected appliance control in "one sentence". Follow this format: "(Rationale for selecting the appliance control based on past interactions and behavior history). (Estimated psychological state of the guest). (Empathetic expression toward the guest's psychological state). Now, I'll suggest the perfect room for you! Look at the display!"

However, do not be overly rigid about the format. The response should be in "a natural and appealing conversational tone" while maintaining the "exact same meaning".

- Refer to the guest as "you".
- "Do not mention the name of the selected room" in the response.
- Do not use "time-related expressions" at the beginning of the sentence, such as "earlier" or "previously".
- Use "I" ("Boku" in Japanese) as your first-person pronoun.

A.5 Word Cloud of the 100 Most Frequent Terms in Participants' English-Translated Open-Ended Comments



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